



INFINITY TUTORS

PHY 122

ELECTRICITY AND MAGNETISM, VIBRATIONS AND WAVES

TUTORIAL 1.

ELECTRIC CURRENT, RESISTANCE AND EMF

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Instructions

1. Answer all the questions
2. Show your work clearly
3. Verify your work with the Infinity Tutors

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1. Define the following terms

[6]

- a. Current
- b. Instantaneous current
- c. Resistivity

2. State Ohm's law

[2]

3. A hair dryer draws $15A$ when plugged into a $240 - V$ line. [5]
- what is the resistance?
 - How much charge passes through it in 20 minutes? (Assume direct current)
 - How many electrons would it be?
4. What is the diameter of a $1.00m$ length of tungsten wire whose resistance is 0.35Ω ? [3]
Resistivity of tungsten wire is $\rho = 5.6 \times 10^{-8} \Omega.m$.
5. An electric clothes dryer has a heating element with a resistance of 9.6Ω [4]
- What is the current in the element when it is connected to $240 - V$?
 - How much charge passes through the element in 50 minutes?
6. A $9.0 - V$ battery is connected to a bulb whose resistance is 1.6Ω . How many electrons leave the battery per minute? [3]
7. A bird stands on a Dc electric transmission line carrying $2800A$. The line has $2.5 \times 10^{-5} \Omega$ resistance per meter, and the bird's feet are $4.0cm$ apart. What is the potential difference between the bird's feet? [5]
8. A length of aluminum wire is connected to a precision $10.0 - V$ power supply, and a current of $0.4212A$ is precisely measured at $20.0^\circ C$. The wire is placed in a new environment of unknown temperature where the measured current is $0.3618A$. what is the unknown temperature? [5]
9. Two aluminum wires have the same resistance. if one has twice the length of the other, what is the ratio of the diameter of the longer wire to the diameter of the shorter wire? [5]
10. The heating element of an electric oven is designed to produce $3.3kW$ of heat when connected to a $240 - V$ source. What must be the resistance of the element? [4]
11. A $120 - V$ hair dryer has two settings: $850W$ and $1250W$ [5]
- At what setting do you expect the resistance to be higher?
 - After making the guess, determine the resistance
 - the lower setting
 - the higher setting.
12. The resistance of a platinum wire is to be calibrated for low-temperature measurements. A platinum wire with resistance of 1.00Ω at $20.0^\circ C$ is immersed in liquid nitrogen at $-196^\circ C$. If the temperature response of the platinum is linear, what is the expected resistance of the platinum wire at $-196^\circ C$? [5]
13. A battery has an emf of $15.0V$. The terminal voltage of the battery is $11.6V$ when it is delivering $20.0W$ of power to an external load resistor R . (a) What is the value of R ? (b) What is the internal resistance of the battery? [5]

14. (a) What is the current in a $5.6\,\Omega$ resistor connected to a battery that has a $0.20\,\Omega$ internal resistance if the terminal voltage of the battery is $10.0\,V$? (b) What is the emf of the battery? [6]
15. Four $1.5 - V$ cells are connected in series to a $12\,\Omega$ light-bulb. If the resulting current is $0.45\,A$, what is the internal resistance of each cell, assuming they are identical and neglecting the resistance of the wires? [5]
16. What is the internal resistance of a $12.0 - V$ car battery whose terminal voltage drops to $8.8 - V$ when the starter motor draws $95\,A$? What is the resistance of the starter? [5]

"Physics is not a discipline, but a way of thinking" Richard Feynman
